

Same Start, Different Weight Spaces: How AdamW and Muon Adapt Dense Retrievers

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Abstract

Does changing the geometry of optimization make a dense retriever use its representation space more effectively? We compare AdamW, Muon, and NorMuon from one pre-trained DenseOn checkpoint on 500,000 shared queries, following 12 configurations through five checkpoints and 14 decontaminated BEIR tasks. [PENDING: the primary all-rate optimizer result and simultaneous interval] We distinguish broader weight-space motion from retrieval-useful dimension allocation by linking reached weights to coordinate-deletion effects and testing prediction at held-out learning-rate doses. A crossed continuation from two fixed reached states then tests whether those states respond differently to further optimization. [PENDING: At the 50K continuation endpoint, the state, operator, and interaction contrasts remain pending. These endpoint contrasts do not decompose the primary-training result.]

1 Introduction

Dense retrieval adapts a pretrained encoder so that a small vector preserves distinctions needed to rank a large corpus. Most adaptation uses AdamW, whose coordinate-wise scaling differs from Muon’s approximately orthogonalized matrix updates (Loshchilov and Hutter, 2019; Liu et al., 2025). Changing that rule changes which weight states the encoder reaches. The central question is whether the resulting motion creates more useful retrieval representations, rather than simply spreading parameter changes across more directions.

There are reasons to expect either outcome. In pretraining studies, Muon learns features with higher effective rank and stronger transfer (Ruan et al., 2026). In contrast, studies of Adam-pretrained models find that switching optimizers during fine-tuning can disrupt learned knowledge, particularly as update strength increases (Qu et al.,

2026; Liu et al., 2026). These findings concern different training settings. Neither tells us how an Adam-pretrained dense retriever’s weight space should change to improve corpus ranking.

Retrieval also exposes a distinction between spreading a representation and using it well. Randomly removing many embedding coordinates can have surprisingly little effect, and removing some coordinates can even improve retrieval (Takeshita et al., 2025). Consequently, a broader spectrum need not mean that more dimensions carry useful ranking evidence. We ask whether optimizer-induced weight changes alter the balance and distribution of helpful and degrading coordinate contributions.

We follow this question from weights to function. Starting from the same DenseOn checkpoint, we train AdamW, Muon, and NorMuon on identical 500,000-query groups with seven hard negatives each. Four rates per optimizer and five checkpoints expose the rate dependence and timing of retrieval changes on 14 decontaminated BEIR tasks. We then trace the reached weight states and measure how coordinate removal changes their representations’ ranking behavior. To distinguish a useful retrieval descriptor from a statistic that merely identifies the optimizer, we test prediction at held-out learning-rate doses.

Figure 1 shows this progression. A separate 2×2 continuation crosses two fixed reached states with reset AdamW or Muon. Calibrating update scale on a common probe lets us ask whether further adaptation depends on both the starting weights and the continuation rule. This intervention concerns whole states and subsequent optimization; it does not isolate any particular geometric feature as the cause of the original retrieval difference.

The study addresses three connected questions:

- when and across which learning rates optimizer choice changes full-corpus retrieval;

From optimizer trajectories to retrieval function

Shared start and training data; compare reached weights, functional dimensions, and retrieval.

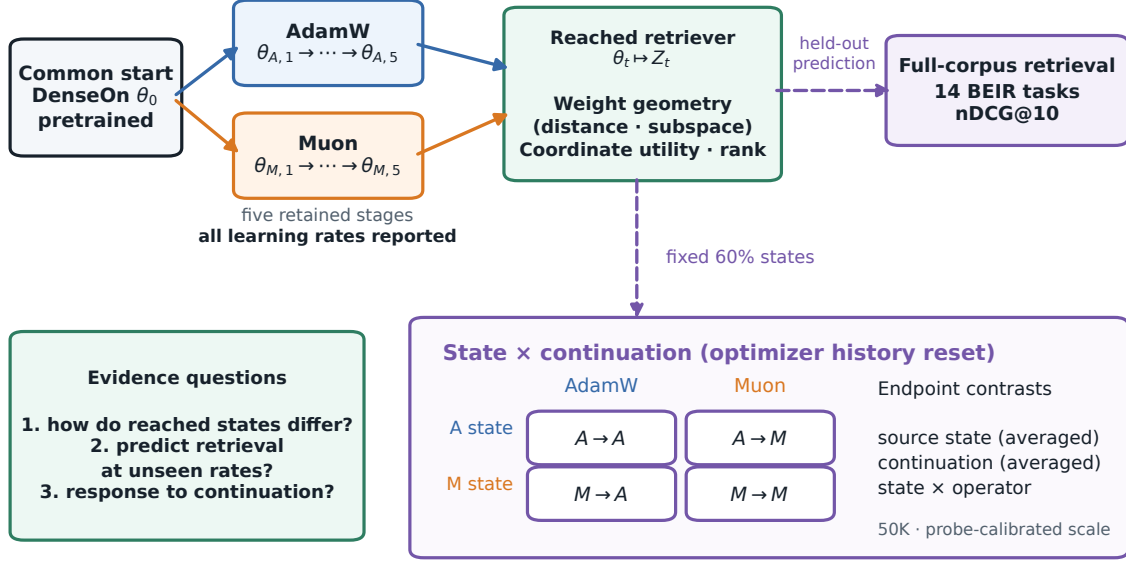


Figure 1: Our explanatory target and evidence chain. Optimizer-specific trajectories create reached weight states and embeddings; weight and coordinate-level measurements become candidate explanations only if they predict full-corpus retrieval out of learning-rate dose. The crossed continuation estimates post-continuation state, operator, and interaction contrasts for one fixed checkpoint pair.

- which differences between reached weight states translate into distributed, helpful coordinate contributions and predict retrieval at held-out doses; and
- whether those reached states respond differently when the continuation optimizer changes.

2 From Optimizer Updates to Retrieval States

2.1 From weight-space motion to ranking evidence

Let $e_\theta(x)$ be the unit-normalized embedding produced by weights θ . The cosine margin between a relevant document d^+ and a competing document d^- is

$$m_\theta(q, d^+, d^-) = s_\theta(q, d^+) - s_\theta(q, d^-) = \sum_{j=1}^{768} e_{\theta,j}(q) [e_{\theta,j}(d^+) - e_{\theta,j}(d^-)]. \quad (1)$$

This identity separates the spread of a representation from its alignment with relevance: different

coordinates may reinforce or oppose a ranking distinction. The encoder maps weight changes into these query–document relationships, so a weight-displacement spectrum alone does not determine whether the changed directions help retrieval.

Our analysis therefore distinguishes three objects: motion in parameter space, the allocation of functional coordinate contributions in the reached representation, and full-corpus ranking. Coordinate deletion measures sensitivity of that representation; because deletion is followed by renormalization, its effects are not the additive terms in Equation 1. The eight-document probe makes these interventions tractable, while the unchanged full-corpus BEIR evaluation supplies the retrieval outcome.

2.2 Update geometry is not an outcome

For a matrix parameter W with gradient G_t , AdamW applies an element-wise second-moment preconditioner. Muon instead applies a Newton–Schulz approximation to the polar factor of a momentum matrix. For the compact SVD $M = U\Sigma V^\top$, its idealized transform is $UV^\top$, which equalizes nonzero singular values (Jordan et al., 2024). NorMuon adds a row-wise historical rescal-

ing after orthogonalization (Li et al., 2025).

Repeated application of these rules changes both weights and the gradients of subsequent updates. We study the accumulated result at saved checkpoints, rather than identify a trajectory with one operator transform. Spectral equalization can emphasize weak directions (Pang et al., 2026), while optimizer switching can disturb pretrained features (Qu et al., 2026). These motivate competing explanations to test through reached states, not an assumption that a flatter spectrum must improve retrieval.

3 Related Work

Dense dual encoders compress queries and documents into single vectors and rank by similarity. BEIR evaluates whether such representations transfer across heterogeneous retrieval tasks (Thakur et al., 2021). DenseOn provides the unsupervised, not-yet-fine-tuned checkpoint and open training recipe used here (Sourty et al., 2026). Representation geometry also matters for retrieval: HIL combines isotropic and anisotropic representations in a late-interaction retriever (Kim et al., 2024). Its emphasis on their balance motivates studying relevance-sensitive consequences of geometric change, rather than maximizing a spread statistic in isolation.

Coordinate ablations provide a more functional view of dimension use. Takeshita et al. (2025) find that text encoders often retain most retrieval quality after random removal of half their dimensions; their tested isotropy, correlation, and outlier measures do not account for the observation, while deleting some individual coordinates improves performance. We use this distinction to ask whether an optimizer merely changes representation rank or changes the distribution of helpful, redundant, and harmful dimensions along its trajectory.

Muon has shown attractive scaling in language-model pretraining, while NorMuon augments its matrix-level transform with neuron-wise adaptation (Liu et al., 2025; Li et al., 2025). Concurrent work warns that switching an Adam-pretrained model to Muon can disrupt pretrained knowledge as update strength grows (Qu et al., 2026; Liu et al., 2026). Other analyses relate Muon to feature rank or margins, identify risks from spectral equalization, or argue for covariance-aware and layer-dependent matrix geometry (Ruan et al., 2026; Pang et al., 2026; Du and Su, 2026; Massena et al., 2026).

Our question is downstream and retrieval-specific: which geometric changes survive optimizer state feedback, preserve unseen scores, and predict corpus ranking?

4 Tracing an Optimizer Through Weight Space

4.1 One controlled adaptation problem

We adapt the released DenseOn unsupervised checkpoint on one deterministic view of 500,000 queries. Each group contains one positive and 7 distinct sampled hard negatives. Queries are allocated proportionally across seven source datasets; a candidate is eligible only when its mined score is below 0.95 times the positive score, and the seven negatives are sampled without replacement from the first ten eligible candidates using stable seed-42 hashes. Every optimizer receives identical group identities and order. We use cosine InfoNCE at temperature 0.02, no in-batch or cross-device negatives, maximum context length 8,192, global batch size 128, one epoch, and bfloat16 forward passes.

The grid contains four learning rates per optimizer: $\{10^{-6}, 3 \times 10^{-6}, 10^{-5}, 3 \times 10^{-5}\}$ for AdamW and $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$ for Muon and NorMuon. Muon-family runs route hidden two-dimensional matrices through the matrix rule and auxiliary parameters through AdamW at 3×10^{-6} . Each run keeps checkpoints at 20, 40, 60, 80, and 100% of 3,907 optimizer steps. Every checkpoint is evaluated on the same 14-task decontaminated BEIR suite.

We use eight NVIDIA L20Z GPUs as two disjoint four-GPU training pools, retaining four ranks per run. Retrieval evaluation is task-parallel across otherwise idle devices and uses exact retrieval rather than approximate ranking. AdamW uses fused CUDA AdamW; Muon and NorMuon use the documented decomposed-bfloat16 Newton-Schulz implementation. All runs retain the same example identities, ordering, checkpoint boundaries, and scoring semantics.

4.2 Comparing rate surfaces and testing retrieval descriptors

The primary estimand averages the four predeclared learning-rate cells within each optimizer at the final checkpoint. This makes the optimizer comparison a statement about the tested rate surface, not a post-hoc best cell. Task-level common resampling supplies simultaneous max- T intervals

Contrast	Mean	95% CI	Decision
Muon – AdamW	–	–	pending
NorMuon – AdamW	–	–	pending
NorMuon – Muon	–	–	pending

Table 1: Final-stage retrieval inference over 14 decontaminated BEIR tasks with simultaneous intervals. Values remain pending until the complete source-bound matrix passes audit.

for the three pairwise contrasts. We also report all five optimizer-by-stage means and a secondary validation-selected comparison, but neither can replace the primary surface-level result.

We test whether a state measurement describes retrieval beyond identifying which optimizer produced it. Nine predeclared weight-space features and four representation-space features are added separately to a baseline containing optimizer, checkpoint stage, and centered log learning rate within optimizer. Four folds each hold out one ordered rate index across all optimizers and stages. Predictive support requires lower pooled held-out RMSE and improvement in at least three folds. The representation features are helpful mass share, normalized helpful participation, degrading mass, and relative shortlist nDCG after removing 50% of coordinates. We report every candidate, including unsupported or unidentified comparisons. This tests generalization to an unseen rate dose, not an unseen retrieval task or causal mediation.

5 Optimizer Effects on Dense Retrieval

The primary estimand averages all four frozen rates within optimizer from a common model, dataset, training schedule, and evaluation suite.

Primary conclusion. [PENDING: the all-rate contrast governs the final optimizer recommendation]

6 How the Optimizers Reshape the Retriever

6.1 Weight-state trajectories

For each of the 88 hidden matrices W_ℓ , we measure the saved-checkpoint segment $W_{\ell,t} - W_{\ell,t-1}$ and cumulative displacement $W_{\ell,t} - W_{\ell,0}$. Magnitude is the joint Frobenius norm of the displacement divided by the joint current weight norm, not an average of matrix-wise ratios. Parameter-weighted summaries describe effective rank, row-norm coefficient of variation, and top-1% row-energy share.

Rank-16 left/right subspace overlap compares all unordered run pairs at each stage; optimizer-pair summaries average all learning-rate pairs equally, without selecting a reference rate.

We retain the original approximate spectral measurements and separately report full-spectrum sensitivity. Entropy rank computed from all singular values is a different quantity from entropy within a renormalized truncated spectrum; the full-spectrum rank summaries explicitly weight nonzero matrices. Neither measurement replaces the other. These are properties of the state produced by many updates, not of a single optimizer step.

[PENDING: the final-stage all-rate hidden-weight displacement and subspace comparison]

6.2 Does Muon use embedding dimensions differently?

Weight motion matters through the representation it creates. The frozen probe contains 16 queries from each of 14 tasks, each with one relevant document and seven fixed competing documents. We measure query/document covariance rank and recompute shortlist retrieval after removing 20 shared random coordinate sets at each of 10, 25, 50, and 75% removal.

The primary attribution uses positive–hardest–negative margin. For task t and coordinate j , let a_{tj} be the task-mean change in margin after deleting j and renormalizing. We average over queries before separating helpful and degrading effects: $h_{tj} = \max(-a_{tj}, 0)$ and $b_{tj} = \max(a_{tj}, 0)$. Writing $H_t = \sum_j h_{tj}$ and $B_t = \sum_j b_{tj}$, the three summaries are

$$\frac{H_t}{H_t + B_t}, \quad \frac{H_t^2}{768 \sum_j h_{tj}^2}, \quad B_t. \quad (2)$$

They measure helpful mass *share*, normalized helpful participation, and degrading mass. Zero denominators are assigned zero by the fixed definition. We compute each summary within task before averaging tasks; analogous shortlist-nDCG attributions are descriptive. These deletion sensitivities are not additive score contributions because renormalization and changes in the hardest competing document couple their effects.

Final-stage optimizer summaries average all four rates. We use 50,000 common task-bootstrap resamples and one fixed-standard-error max- T family over the nine combinations of three features and three optimizer contrasts. More constructive coordinate use requires all three simultaneous 95%

intervals in the desired directions: greater helpful share and participation, less degrading mass. Failure of this joint criterion does not establish equivalence.

Coordinate attribution is basis dependent. We repeat the endpoint analysis after three shared Haar rotations, each of which preserves all full-vector cosine scores. A native-coordinate difference can support a claim about native-coordinate utilization. Directional agreement across the three tested rotations is a robustness check, not proof of invariance under arbitrary bases or of greater dimensional capacity.

[PENDING: the all-rate dimension-utilization comparison, random-removal curves, and shared-rotation control]

6.3 Which state measurements predict retrieval?

Every weight- and representation-space candidate faces the same leave-dose-index-out test. The baseline already knows optimizer identity, checkpoint stage, and within-optimizer dose, so a feature must explain variation beyond recognizing Muon. We report all candidates, including unsupported ones, and treat predictive support as a bridge to retrieval rather than causal mediation.

[PENDING: the leave-dose-index-out weight-geometry-to-retrieval bridge]

[PENDING: the leave-dose-index-out dimension-to-retrieval bridge]

7 How Do Reached States Respond to Further Optimization?

Retrieval at a fixed checkpoint depends on its weights, not on an optimizer applied only later. Our follow-up therefore asks a conditional question: does the reached state change the response to further optimization? We fix the 60% AdamW 3×10^{-5} and Muon 3×10^{-4} checkpoints, reset optimizer history, and continue each with reset AdamW and reset Muon. The source pair is fixed, not selected from the new BEIR outcomes or averaged over the primary rate surface.

Each cell’s hidden learning rate is calibrated to a global hidden update-to-weight Frobenius ratio of 5×10^{-4} on the fixed 32-example gradient-history probe, then held fixed as the schedule’s base rate. This does not match the first realized shuffled-batch update or subsequent update norms. Both rules use the same hidden/auxiliary partition and auxiliary

AdamW at 3×10^{-6} ; calibration excludes weight decay. The four cells share each of three fixed data orders and a 50,000-query horizon.

For each seed–task pair, let Y_{so} be final nDCG@10 after this continuation, with source $s \in \{A, M\}$ and continuation rule $o \in \{A, M\}$. The frozen contrasts are

$$\Delta_S = \frac{1}{2}[(Y_{MA} - Y_{AA}) + (Y_{MM} - Y_{AM})], \quad (3)$$

$$\Delta_O = \frac{1}{2}[(Y_{AM} - Y_{AA}) + (Y_{MM} - Y_{MA})], \quad (4)$$

$$\Delta_I = (Y_{MM} - Y_{MA}) - (Y_{AM} - Y_{AA}). \quad (5)$$

They concern post-continuation endpoints, not gains relative to each source checkpoint. In particular, $Y_{MM} - Y_{AA} = \Delta_S + \Delta_O$; the interaction is not a third additive component. A positive averaged contrast need not hold within both levels, and positive interaction can mean less harm rather than improvement. Each contrast receives its own 95% two-way seed/task bootstrap interval from 100,000 resamples; these are marginal, not simultaneous intervals. An inconclusive contrast is not equivalent to zero or dominated by one whose interval excludes zero.

[PENDING: The averaged source-state contrast, averaged continuation-operator contrast, and state-operator interaction remain pending, including all three marginal intervals and decisions. Positive interaction alone would not establish beneficial continuation or full-trajectory co-adaptation.]

8 Discussion

Optimizer quality is a property of a trajectory surface. The all-rate estimand prevents the argument from collapsing onto one fortunate learning rate, while five checkpoint stages reveal whether an advantage emerges early, accumulates gradually, or disappears. The result remains specific to the tested DenseOn adaptation problem; it is stronger than a best-cell comparison and weaker than a universal optimizer ranking.

Weight-space separation and functional change need not coincide. Muon can produce spectrally distinct displacements without improving coordinate utility. Conversely, small changes in weights can alter which query–document distinctions a representation preserves. The held-out-dose tests ask which measurements track corpus ranking beyond optimizer identity, stage, and dose. If none adds predictive value, the tested weight-space explanation remains unresolved even when retrieval differs; visual separation cannot supply the missing link.

[PENDING: all-rate random-removal curves and five-stage coordinate-utility trajectories from the complete primary matrix; no results are plotted in this development placeholder]

Figure 2: Reserved layout for the complete primary dimension-utility figure.

Dimension count and dimension utility are different. Random-removal robustness can arise from redundancy, distributed coding, or cancellation between helpful and degrading coordinates. Effective rank can also disagree with the functional ablation readout. The coordinate analysis therefore asks which parts of the representation help retrieval, how distributed that utility is, and whether any optimizer difference survives a change of basis.

Continuation tests state-dependent responses, not a gain decomposition. The crossed design asks whether a source-state advantage persists on average after further adaptation and whether the relative continuation preference changes between the fixed states. Neither an averaged main effect nor an interaction alone establishes benefit within both states, additivity, or full-trajectory co-adaptation. Explaining a particular geometric feature causally would require an intervention on that feature, beyond the present state-level comparison.

9 Conclusion

[PENDING: the primary all-rate optimizer result and simultaneous interval]

[PENDING: the conclusion supported by the dimension-utilization and rotation analyses]

[PENDING: The crossed 50K continuation findings remain pending. The design resets optimizer state and matches the fixed calibration-probe scale; its endpoint contrasts do not decompose the primary-training result or establish the causal role of a geometric feature.]

The central distinction is between distributing weight-space motion and distributing useful ranking evidence. Reached-weight measurements, coordinate interventions, held-out-dose prediction, and conditional continuation provide complementary views of that distinction. Together they can narrow an explanation of optimizer-dependent retrieval without treating an intrinsic update property as a new discovery or claiming a causal decomposition of the original gain.

10 Limitations

Primary inference concerns one 149M-parameter DenseOn checkpoint, one English supervised corpus, and one epoch of full-parameter adaptation. The base model was pretrained with an Adam-

family optimizer, so this is a study of optimizer switching rather than Muon-from-scratch pretraining. Four learning rates provide a broad but finite grid. The all-rate contrast reduces best-cell selection dependence but remains a one-seed, pinned-grid result; the secondary selected-recipe comparison does not replace replication across training views.

Decontamination reduces known benchmark overlap but cannot establish the absence of every semantic duplicate. Aggregate MTEB outputs lack per-query rankings, so uncertainty is computed over tasks and we make no query-level BEIR significance claim. The geometry and dimension bridges are predictive rather than causal and may miss useful properties outside the frozen candidates. Coordinate ablation uses an eight-document probe, and coordinate identities are not invariant under rotation; the rotation control limits, rather than eliminates, this dependence.

The crossed reset-state factorial concerns one fixed 60% checkpoint pair and a 50K-query continuation horizon. Its three order seeds do not replicate source-model training, and its marginal intervals are not family-wise controlled. It changes entire source states and continuation rules, not a measured geometry feature, and therefore does not identify mediation over the full 500K trajectory or establish why a feature predicts retrieval.

11 Ethical Considerations

This study does not collect new human-subject data or deploy a retrieval system to users. Released models and corpora can nevertheless contain social biases, harmful material, copyrighted text, or personal information. Higher benchmark nDCG is not evidence of fairness, safety, privacy, or deployment readiness. A production retriever requires corpus-specific auditing, filtering, access controls, and human review appropriate to its users and failure costs.

We report useful wall time, throughput, GPU-hours, and optimizer-state size because an optimizer recommendation has resource consequences. These measurements are specific to the pinned hardware and software stack and do not justify unbounded hyperparameter search. Public artifacts must respect the licenses and usage terms of upstream models, datasets, and evaluation collections.

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A Artifact and Reproducibility

We preserve the exact model and dataset revisions, deterministic 500K training ledger, negative sampling seeds, optimizer routing, package locks, W&B histories, checkpoint hashes, and evaluation receipts. Every reported table is generated only after source-bound completeness and provenance audits. The public artifact manifest binds each model state and result to its byte count and digest; the renderer fails closed when an expected run, checkpoint, task, or source identity differs.

B Complete Weight-Space and Dimension Readouts

The main text reports the weight- and representation-space findings that bear on the mechanism. For completeness, the tables below retain every frozen candidate, including negative results.

C State-by-Operator Mechanism Test

The crossed experiment resets optimizer state at both source checkpoints, matches the fixed calibration-probe update scale, and reports all three co-primary contrasts regardless of sign. Exact seed–task contrasts and all five probe stages remain in the source-addressed artifact tables.

Frozen geometry feature	ΔRMSE	Folds	Supported	Residual ρ
log segment/weight norm	–	–/4	pending	–
segment stable-rank fraction	–	–/4	pending	–
segment effective-rank fraction	–	–/4	pending	–
segment row-norm CV	–	–/4	pending	–
segment top-1% row energy	–	–/4	pending	–
cumulative/weight norm	–	–/4	pending	–
cumulative stable-rank fraction	–	–/4	pending	–
segment subspace overlap to AdamW	–	–/4	pending	–
cumulative subspace overlap to AdamW	–	–/4	pending	–

Table 2: Four-fold leave-dose-index-out geometry-to-retrieval bridge. Positive ΔRMSE means the feature lowers pooled held-out error. Values remain pending.

Dimension-use feature	Contrast	Effect	Simultaneous 95% CI	Decision
Helpful mass share	Muon–AdamW	–	–	pending
Helpful participation	Muon–AdamW	–	–	pending
Degrading mass	Muon–AdamW	–	–	pending
Helpful mass share	NorMuon–AdamW	–	–	pending
Helpful participation	NorMuon–AdamW	–	–	pending
Degrading mass	NorMuon–AdamW	–	–	pending

Table 3: Final-stage, all-rate dimension-utilization contrasts against AdamW. The common max- T family also includes the corresponding NorMuon–Muon contrasts; values remain pending until the complete source-bound analysis passes audit.

Factorial contrast	Effect	Marginal 95% CI	Decision
Muon-state – AdamW-state	–	–	pending
Muon operator – AdamW operator	–	–	pending
State $\times$ operator interaction	–	–	pending

Table 4: Prospective reset-state factorial. All results remain pending. The three marginal intervals do not provide simultaneous family-wise coverage.

D Interpretation of the Analyses

E Full Experimental Matrix

The DenseOn matrix contains three optimizers, four rates per optimizer, five checkpoints, and 14 tasks: 12 runs, 60 checkpoints, and 840 checkpoint-task units. The released result tables additionally contain every optimizer-by-stage mean, validation-selected secondary contrast, per-task final score, wall time, throughput, peak allocation, and checkpoint size. No run is removed on the basis of retrieval quality.

Claim	Required evidence	Insufficient evidence alone
Faster adaptation	useful wall time, examples/s, GPU-hours, matched-hardware time-to-quality	lower loss at equal steps
Better DenseOn optimizer	all-rate contrast over 14 tasks	best BEIR cell alone
Weight motion explains retrieval	held-out prediction beyond optimizer, stage, and dose	larger displacement alone
Dimensions are used more constructively	helpful mass share and normalized participation rise while degrading mass falls, with simultaneous intervals	effective rank alone
Geometry predicts retrieval	held-out prediction beyond optimizer, stage, and dose	optimizer-identifying geometry
Source-state advantage persists on average after continuation	crossed endpoint state contrast with calibrated probe scale	one-step functional proxy
Continuation preference depends on reached state	crossed endpoint interaction	full-trajectory co-adaptation

Table 5: How the analyses distinguish retrieval outcomes, predictive descriptors, and conditional continuation effects. Each comparison has a different scope of interpretation.